

Style: layout choices Based on design principles

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Style: Layout Choices

Based on design principles

This guide provides reasons that some choices in layout design are better than others.

The two samples below show two ways to arrange data around a figure. The sample on the left is a common way among student engineers; the sample on the right shows an arrangement used by many journals.

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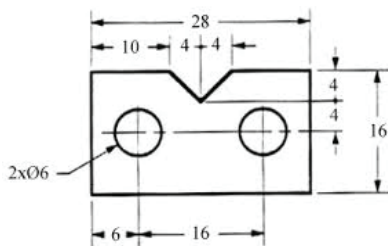
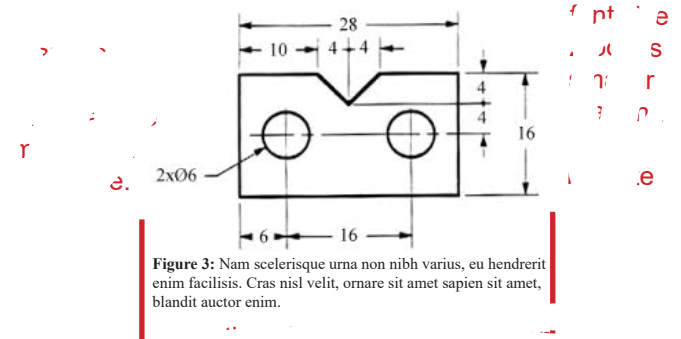


Figure 3: Nam scelerisque urna non nibh varius, eu hendrerit enim facilisis. Cras nisl velit, ornare sit amet sapien sit amet, blandit auctor enim.

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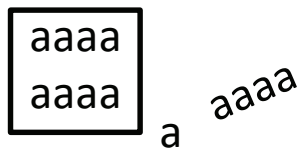
The sample on the right arranges the data better than the sample on the left because items are rearranged spatially to better show relationships. At right, the caption now is closer to the figure than it was at left, and its font size is smaller (2 points) than the font size of the body text. Finally, the margins of the caption at right are the same as the margins of the figure. These three changes in layout at right link the caption to the figure more clearly than the layout shown in the sample at left.

This guide covers five design principles and shows their application in common engineering documents such as a lab report and an IEEE journal article.

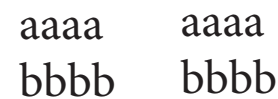
Applying these five principles can improve the visual style of documents and help readers.

Five principles of document design

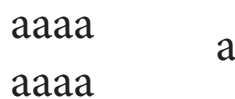
1. Closure: a closed shape defines a group.



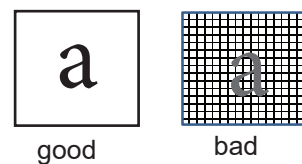
4. Symmetry: balanced items seem complete.



2. Proximity: items close together appear related.



5. Figure/Ground segregation: a figure should be distinct from its ground.



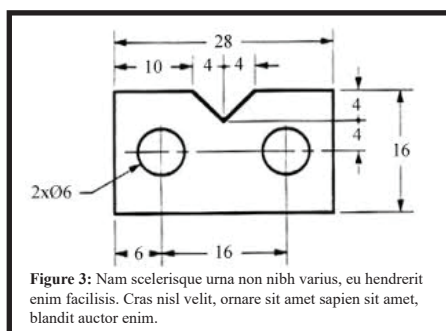
3. Similarity: items that look alike will be grouped together.



Common applications in documents

1. Closure: A common problem is redundant closure whereby a box is **not** needed because white space already provides the necessary closure separating the figure/caption from the body text. Sometimes a box may be needed for emphasis or to show closure, but if not, then the box may be omitted as shown below, right.

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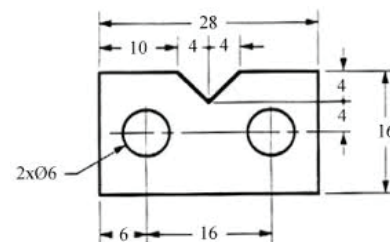


Figure 3: Nam scelerisque urna non nibh varius, eu hendrerit enim facilisis. Cras nisl velit, ornare sit amet sapien sit amet, blandit auctor enim.

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2 & 3. Proximity and Similarity: the samples from the previous page are repeated below with the added labels of proximity and similarity that identify these two principles in the improved sample at right.

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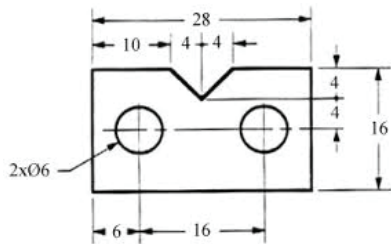


Figure 3: Nam scelerisque urna non nibh varius, eu hendrerit enim facilisis. Cras nisl velit, ornare sit amet sapien sit amet, blandit auctor enim.

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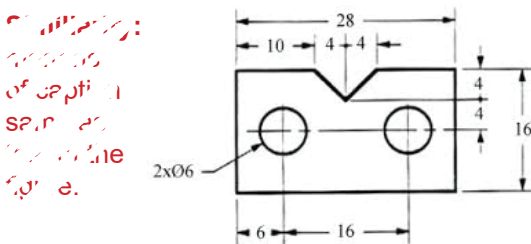


Figure 3: Nam scelerisque urna non nibh varius, eu hendrerit enim facilisis. Cras nisl velit, ornare sit amet sapien sit amet, blandit auctor enim.

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4. Symmetry (balance): Balanced titles.

A Competitive Robot ... Playing Air Hockey with Human Beings

A Competitive Robot for Playing Air Hockey with Human Beings

These five design principles often overlap. In the sample at left, the unit “human beings” is split onto the next line, thus violating closure and proximity. The revision at right thus corrects three principles: symmetry, closure, and proximity.

5. Figure/Ground segregation

Shading Can Often Cause Problems

Shading Can Often Cause Problems

The sample at left shows a common problem among engineers: the temptation to over-design. Many options are available to embellish text and visuals, yet some options distract from the content and from good readability. At left, more ink is given to the design than to the data.

6. Design principles in context: The spread on the next two pages shows these design principles in context. Some engineering professors prescribe the IEEE format for students’ final reports. The left-facing page shows violations of the principles (with commentary) on the first page of an IEEE-style report, and the right-facing page shows the corrections.

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First A. Author, Second B. Author, Third C. Author, Fourth D.

Author

Abstract—Duis a auctor augue. Vivamus in pretium risus. Curabitur sit amet vulputate mi, eu mattis urna. Fusce augue magna, bibendum rhoncus tincidunt eget, suscipit vitae mi. Fusce augue nibh, eleifend in erat id, dapibus congue libero. Aliquam vel aliquam neque, id cursus nibh. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Fusce vitae facilisis odio, ut auctor sem. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Maecenas suscipit ex a cursus semper. Sed consequat dolor quis aliquam pharetra. Aliquam sagittis quam et pulvinar interdum. Morbi fermentum enim nec tempor feugiat. Morbi mi erat, cursus posuere gravida ac, cursus at ligula. Nam mattis aliquam lacus nec ultricies. Orci varius natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Cras ac purus mattis, tristique sapien eget, aliquet justo. Nullam eget cursus diam, eu pellentesque sem. Donec eros ante, scelerisque eget leo at, euismod pharetra lacus.

Index Terms—Abacus, Botticelli, Caiphaz, Demothenes, Euripedes, Franconian, Grancel, Holothenes, Iphigenia.

I. INTRODUCTION

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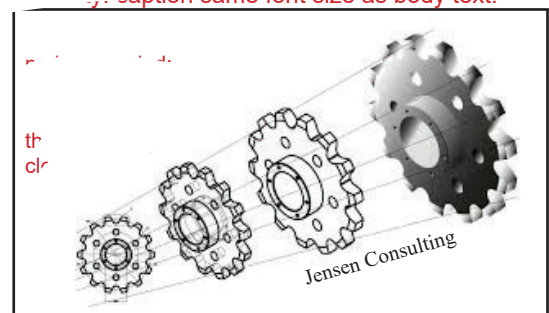
II. FUSCE VITAE FACILISIS ODIO

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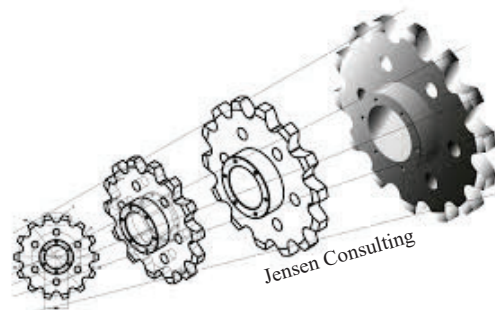


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Comments

1. A common feature in student texts is unequal spacing among the words, known as bad tracking. It is caused in Microsoft products when authors justify the right margins to have the text look professional. This phenomenon is more frequent in narrow columns of text, especially in two-column text.

Writers can correct bad tracking by setting the right margin as ragged (shown at right) or by using professional-grade software such as Adobe InDesign.

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setting a ragged right margin

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tur hendrerit fringilla dui, id convallis diam commodo ac. Integer
maximus dui orci, eget semper risus vehicula nec.

2. The five design principles are based on “Gestalt Theory and In-
structional Design” by Patrick Moore and Chad Fitz, *JTWC* vol.
23(2) 137-157, 1993.